

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	2 Marks

SMART METRO TICKET GENERATING SYSTEM

Problem – Solution Fit

1. Problem Statement

Metro commuters today rely on a largely manual, paper-based ticketing process: tokens or paper tickets are issued at staffed counters, payment is predominantly cash, and gate checks are performed manually by station staff. As ridership grows, this process creates friction at every stage of the commuter journey and leaves operators without the data they need to run the network efficiently.

These pain points were gathered through direct station observation during peak and off-peak hours, structured interviews with station staff and Metro Operations, and a review of existing revenue-reconciliation and complaint logs. Each problem below is stated with the evidence that substantiates it, so that the resulting solution can be evaluated against real operating conditions rather than assumptions.

Five problems, gathered from station observation and operator interviews, define the scope this solution must address:

- Long queues at ticket counters during morning and evening peak hours.
- Paper tickets that are easily lost, damaged, or counterfeited.
- Cash-heavy transactions that slow daily revenue reconciliation.
- Fare evasion enabled by manual, inconsistent gate checks.
- Lack of real-time ridership and revenue data for operational decisions.

2. Mapping Problems to Solution Capabilities

Each identified problem is mapped directly to a specific capability of the Smart Metro Ticket Generating System, ensuring that every feature in the proposed solution exists to address a real, evidenced pain point rather than being built speculatively. Problems were also scored on how often they occur and how much they impact commuters and operations, so the team could prioritize which capabilities to deliver first in the pilot phase.

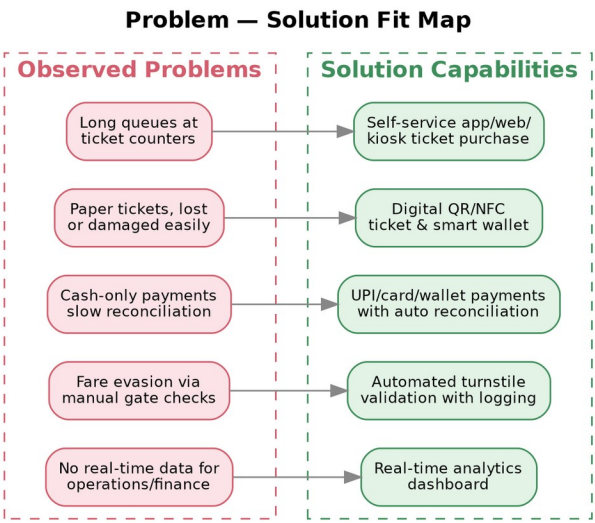


Figure 1: Problem-to-solution capability mapping

2.1 Problem Severity Matrix

Plotting each problem by frequency of occurrence against business/commuter impact shows that fare evasion and cash-heavy reconciliation sit in the highest-priority quadrant, confirming why automated gate validation and digital payments were prioritized as Phase 1 pilot capabilities.

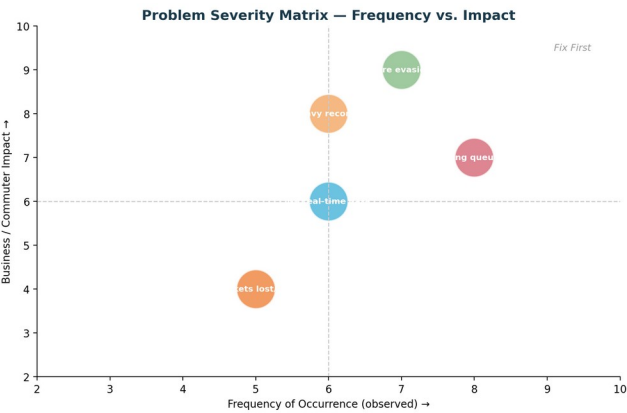


Figure 2: Problem severity matrix — frequency of occurrence vs. business/commuter impact

2.2 Fit Evidence Table

Problem	Evidence	Solution Fit
Long queues at ticket counters	Avg. wait time ~90 sec at peak stations	Self-service app/web/kiosk purchase
Paper tickets lost/damaged	Manual reissue requests at counters	Digital QR/NFC ticket & smart wallet
Cash-heavy, slow reconciliation	~12 hrs manual daily reconciliation	Digital payments with auto reconciliation
Fare evasion at manual gates	~8% estimated fare leakage	Automated turnstile validation + audit log
No real-time operational data	Reports compiled manually, day+ lag	Real-time analytics dashboard

3. Expected Impact

The chart below compares baseline current-state metrics against the targets the proposed solution is expected to achieve once fully deployed. These figures form the basis for measuring solution fit after pilot rollout: if actual pilot results track close to these targets, the fit between problem and solution is considered validated.

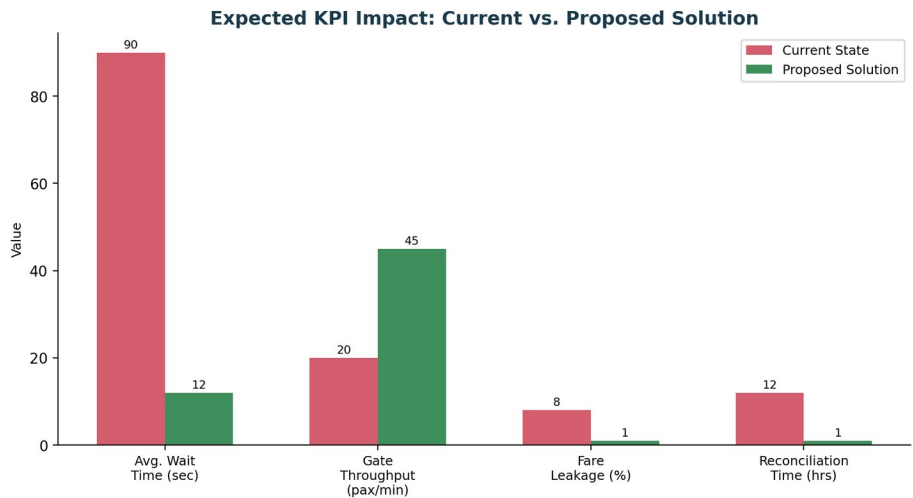


Figure 3: Expected KPI impact — current state versus proposed solution targets

3.1 Validating the Fit

- Wait time and gate throughput will be measured directly from turnstile logs during the pilot phase.
- Fare leakage will be estimated by comparing ticket-validation counts against station entry/exit counts.
- Reconciliation time will be measured from transaction close to finance sign-off in the new digital workflow.
- Any metric that does not trend toward its target within the pilot window will trigger a review of the corresponding feature before regional rollout.

3.2 Conclusion

The problem–solution fit analysis shows a direct, evidence-backed line from each operational pain point to a specific capability of the Smart Metro Ticket Generating System. This tight mapping — rather than a generic technology upgrade — is what justifies moving from problem validation into the proposed solution and, subsequently, the solution architecture defined in the companion documents.